Q3(c): Risk Aversion and Marginal Utility
Understanding u' is Declining

ECO 310

PART 1: FOUNDATIONS

What is a Utility Function?

Utility Function: $u(w)$
A mathematical function that represents how much **satisfaction or happiness** a person gets from having wealth w.

Think of it like a happiness meter:

- More wealth $\Rightarrow$ higher utility (you're happier)
- But: each extra dollar might not make you *as much* happier as the previous one

Example Utility Functions

1. Square Root: $u(w) = \sqrt{w}$

- $w = 100 \Rightarrow u(100) = 10$
- $w = 400 \Rightarrow u(400) = 20$
- $w = 900 \Rightarrow u(900) = 30$

2. Logarithm: $u(w) = \ln(w)$

- $w = 10 \Rightarrow u(10) = 2.30$
- $w = 100 \Rightarrow u(100) = 4.61$
- $w = 1000 \Rightarrow u(1000) = 6.91$

3. Linear: $u(w) = w$

- $w = 100 \Rightarrow u(100) = 100$
- $w = 200 \Rightarrow u(200) = 200$

What is Marginal Utility? (Understanding u')

Marginal Utility: $u'(w) = \frac{du}{dw}$
The **rate of change** of utility with respect to wealth.
Translation: How much does my happiness increase if I get one more dollar?

Understanding Derivatives Intuitively

The derivative $u'(w)$ tells you the **slope** of the utility function at wealth level w.

High slope (steep): Each extra dollar gives you a lot more utility

Low slope (flat): Each extra dollar gives you only a little more utility

Example 1: $u(w) = \sqrt{w}$

Take the derivative:

$$u'(w) = \frac{d}{dw}\sqrt{w} = \frac{1}{2\sqrt{w}}$$

Let's calculate at different wealth levels:

Wealth w	**Utility** $u(w)$	**Marginal Utility** $u'(w)$
1	1	0.5
100	10	0.05
10000	100	0.005

Key observation: As wealth increases, marginal utility **decreases**!

When you're poor ($w = 1$), an extra dollar is worth 0.5 units of utility.

When you're rich ($w = 10000$), an extra dollar is only worth 0.005 units of utility.

Example 2: $u(w) = \ln(w)$

Take the derivative:

$$u'(w) = \frac{d}{dw}\ln(w) = \frac{1}{w}$$

Wealth w	**Utility** $u(w)$	**Marginal Utility** $u'(w)$
1	0	1
10	2.30	0.1
100	4.61	0.01

Again: marginal utility **decreases** as wealth increases!

What is the Second Derivative? (Understanding u'')

Second Derivative: $u''(w) = \frac{d^2u}{dw^2} = \frac{d(u')}{dw}$
The **rate of change of marginal utility**.
Translation: How fast is marginal utility decreasing (or increasing)?

What Does u'' Tell Us?

If $u'' < 0$: Marginal utility is **decreasing** (utility function curves downward)

This is called **CONCAVE**

If $u'' > 0$: Marginal utility is **increasing** (utility function curves upward)

This is called **CONVEX**

If $u'' = 0$: Marginal utility is **constant** (utility function is a straight line)

This is called **LINEAR**

Example: $u(w) = \sqrt{w}$

First derivative: $u'(w) = \frac{1}{2\sqrt{w}}$

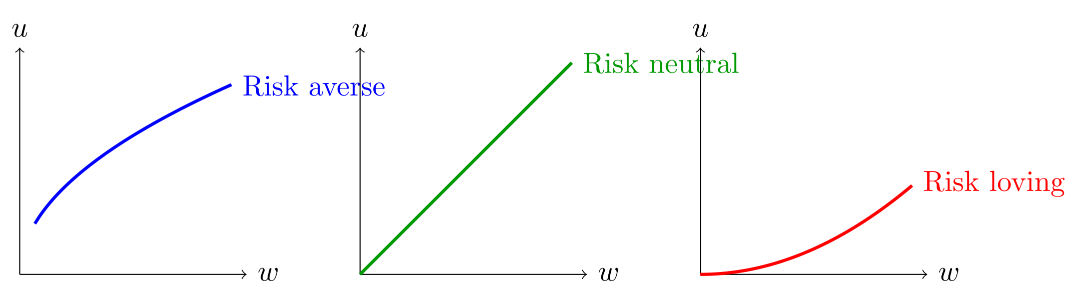
Second derivative:

$$u''(w) = \frac{d}{dw}\left(\frac{1}{2\sqrt{w}}\right) = \frac{d}{dw}\left(\frac{1}{2}w^{-1/2}\right) = -\frac{1}{4}w^{-3/2} = -\frac{1}{4w^{3/2}}$$

Notice: $u''(w) < 0$ for all $w > 0$!

This means marginal utility is always decreasing.

Visual: Concave vs Convex vs Linear



Concave ($u'' < 0$) Linear ($u'' = 0$) Convex ($u'' > 0$)

What is Risk Aversion?

Risk Aversion: A person is risk averse if $u''(w) < 0$
This means: **They prefer a sure thing over a gamble with the same expected value.**

Example: Risk Averse Person

Suppose $u(w) = \sqrt{w}$ and current wealth is $w = 100$.

Option A (Sure thing): Keep \$100 for sure

$$u(100) = \sqrt{100} = 10$$

Option B (Gamble): 50% chance of \$0, 50% chance of \$200

$$\text{Expected utility} = 0.5 \cdot u(0) + 0.5 \cdot u(200) = 0.5(0) + 0.5(14.14) = 7.07$$

Expected wealth is the same:

- Option A: \$100 for sure
- Option B: $0.5(0) + 0.5(200) = \$100$ expected

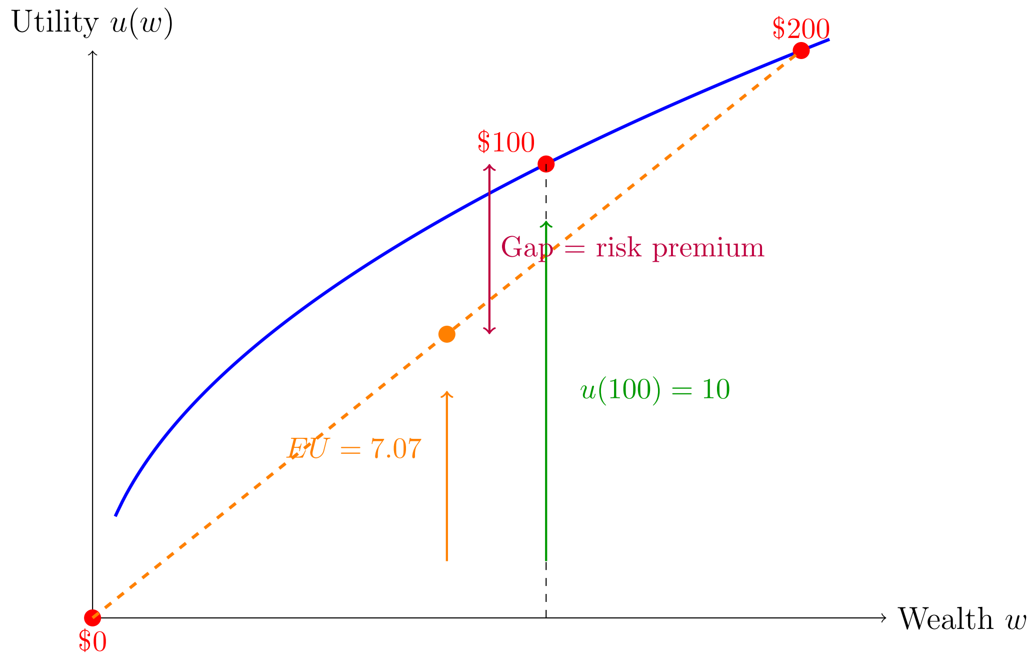
But utility is different:

- Option A: $u = 10$
- Option B: $u = 7.07$

Conclusion: Risk averse person prefers Option A (sure thing) even though both have same expected wealth!

This happens because utility function is **concave** ($u'' < 0$).

Visual Explanation



Key insight: The concave curve means the utility of the average is **higher** than the average of the utilities!

Absolute Risk Aversion (ARA)

Absolute Risk Aversion (ARA):

$$A(w) = -\frac{u''(w)}{u'(w)}$$

Measures **how risk averse** you are in absolute terms.

What Does ARA Tell Us?

Higher ARA: More risk averse (avoid risk more)

Lower ARA: Less risk averse (willing to take more risk)

Why the negative sign? Because $u'' < 0$ for risk averse people, so the negative makes $A(w) > 0$.

CARA: Constant Absolute Risk Aversion

If $A(w)$ is **constant** (doesn't change with wealth), we call this CARA.

Example: $u(w) = -e^{-\alpha w}$ where $\alpha > 0$

Calculate:

$$u'(w) = \alpha e^{-\alpha w}$$

$$u''(w) = -\alpha^2 e^{-\alpha w}$$

$$A(w) = -\frac{-\alpha^2 e^{-\alpha w}}{\alpha e^{-\alpha w}} = \alpha$$

ARA is constant! No matter your wealth, you have the same absolute risk aversion.

DARA: Decreasing Absolute Risk Aversion

If $A(w)$ **decreases** as wealth increases, we call this DARA.

Intuition: As you get richer, you're willing to take more absolute risk (invest more dollars in risky assets).

Example: $u(w) = \sqrt{w}$

Calculate:

$$u'(w) = \frac{1}{2\sqrt{w}}$$

$$u''(w) = -\frac{1}{4w^{3/2}}$$

$$A(w) = -\frac{-\frac{1}{4w^{3/2}}}{\frac{1}{2\sqrt{w}}} = \frac{1}{4w^{3/2}} \cdot \frac{2\sqrt{w}}{1} = \frac{1}{2w}$$

As w increases, $A(w)$ decreases! This is DARA.

Relative Risk Aversion (RRA)

Relative Risk Aversion (RRA):

$$R(w) = -\frac{w \cdot u''(w)}{u'(w)} = w \cdot A(w)$$

Measures **how risk averse** you are in **relative** (percentage) terms.

What's the Difference Between ARA and RRA?

Absolute Risk Aversion (ARA): How many **dollars** are you willing to risk?

Relative Risk Aversion (RRA): What **fraction of your wealth** are you willing to risk?

Example: ARA vs RRA

Suppose you have two people:

- Person A: wealth = \$10,000
- Person B: wealth = \$1,000,000

Both have same RRA = 2, but different ARA.

Question: How much would each invest in a risky asset?

If RRA is constant, they invest the **same percentage** of wealth:

- Person A might invest 50% = \$5,000
- Person B might invest 50% = \$500,000

Even though Person B invests way more **dollars** (higher absolute risk), they both invest the same **fraction** of wealth (same relative risk).

CRRA: Constant Relative Risk Aversion

If $R(w)$ is **constant**, we call this CRRA.

Example: $u(w) = \frac{w^{1-\gamma}}{1-\gamma}$ where $\gamma > 0$ (and $\gamma \neq 1$)

Calculate:

$$u'(w) = w^{-\gamma}$$
$$u''(w) = -\gamma w^{-\gamma-1}$$
$$R(w) = -\frac{w \cdot (-\gamma w^{-\gamma-1})}{w^{-\gamma}} = \frac{\gamma w^{-\gamma}}{w^{-\gamma}} = \gamma$$

RRA is constant! No matter your wealth, you invest the same **fraction** in risky assets.

Special Case: $u(w) = \ln(w)$

This is CRRA with $\gamma = 1$:

$$u'(w) = \frac{1}{w}$$

$$u''(w) = -\frac{1}{w^2}$$

$$R(w) = -\frac{w \cdot (-1/w^2)}{1/w} = \frac{1/w}{1/w} = 1$$

RRA = 1 (constant)

Summary Table: Risk Aversion Measures

Measure	**Formula**	**Interpretation**
Risk aversion	$u''(w) < 0$	Concave utility, prefer certainty
ARA	$A(w) = -\frac{u''(w)}{u'(w)}$	Absolute risk aversion
RRA	$R(w) = -\frac{w \cdot u''(w)}{u'(w)}$	Relative risk aversion
CARA	$A(w) = \text{constant}$	Same absolute risk at all wealth
DARA	$A'(w) < 0$	Less absolute risk averse when richer
CRRA	$R(w) = \text{constant}$	Same % risk at all wealth

Common Utility Functions

Function	$u(w)$	**ARA**	**RRA**
Logarithmic	$\ln(w)$	$\frac{1}{w}$ (DARA)	1 (CRRA)
Power	$\frac{w^{1-\gamma}}{1-\gamma}$	$\frac{\gamma}{w}$ (DARA)	γ (CRRA)
Square root	$\sqrt{w}$	$\frac{1}{2w}$ (DARA)	$\frac{1}{2}$ (CRRA)
Exponential	$-e^{-\alpha w}$	α (CARA)	αw (IRRA)
Linear	w	0	0 (Risk neutral)

Note: IRRA = Increasing Relative Risk Aversion

PART 2: APPLICATION TO Q3(c)

What is Part (c) Asking?

Q3(c) asks: *"Give an interpretation for why the DM prefers state C to state A when the condition from part (b) holds."*

From part (b), the condition at an interior optimum is:

$$\frac{u'(w^C)}{u'(w^A)} = \frac{r_1^A - r_2^A}{r_2^C - r_1^C} \qquad (*)$$

The question wants us to interpret: **When does the DM prefer state C over state A?**

What Does "Prefer State C" Mean?

The DM prefers state C if they have **higher wealth** in state C than in state A:

$$w^C > w^A$$

where:

- $w^A = w_0 + r_1^A x + r_2^A(100 - x)$ (wealth in state A)
- $w^C = w_0 + r_1^C x + r_2^C(100 - x)$ (wealth in state C)

Key insight: More wealth = higher utility (since $u' > 0$).

So: $w^C > w^A \Rightarrow u(w^C) > u(w^A)$

What Does u' Mean?

Notation:
$u'(w) = \frac{du}{dw}$ = marginal utility of wealth
This measures: "How much does utility increase per additional dollar?"

Example:

If $u(w) = \sqrt{w}$, then:

$$u'(w) = \frac{1}{2\sqrt{w}}$$

When $w = 100$: $u'(100) = \frac{1}{20} = 0.05$

When $w = 10000$: $u'(10000) = \frac{1}{200} = 0.005$

Notice: As wealth increases, marginal utility **decreases**.

What Does "Risk Aversion" Mean?

Risk Aversion $\Leftrightarrow u''(w) < 0$
The utility function is **concave** (curves downward).

Three Key Facts:

1. $u'(w) > 0$: More wealth is better (increasing function)
2. $u''(w) < 0$: Marginal utility is decreasing (concave function)
3. **Risk aversion** means: diminishing marginal utility of wealth

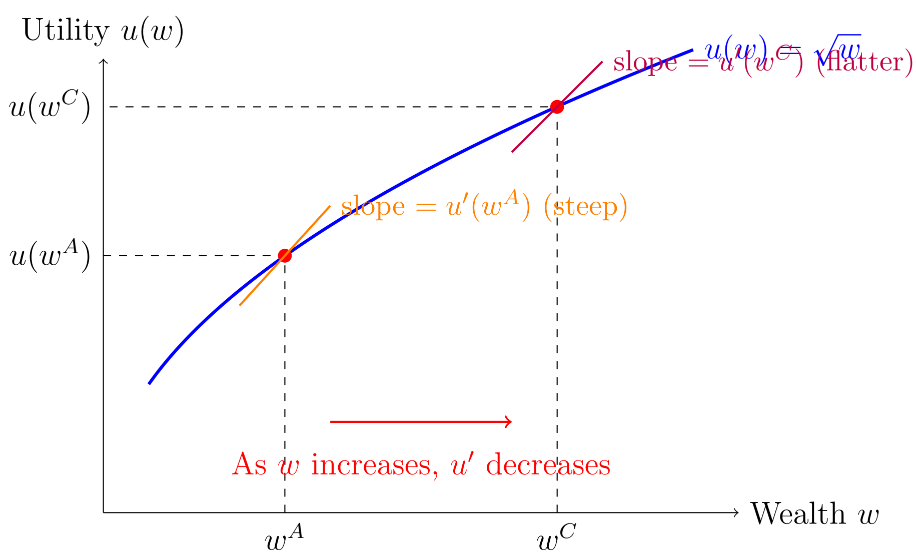
What is u''?

$$u''(w) = \frac{d^2u}{dw^2} = \frac{d(u')}{dw}$$

This is the **rate of change of marginal utility**.

If $u'' < 0$: Then u' is a **decreasing function** of w.

Visual Explanation of Risk Aversion



Key observation: The slope gets **flatter** as wealth increases.

$$w^C > w^A \quad \Rightarrow \quad u'(w^C) < u'(w^A)$$

The Logic Chain

The solution says: "He prefers state C if $w^C > w^A$; by risk aversion, u' is declining so this holds iff $u'(w^C) < u'(w^A)$."

Let's break this down:

Step 1: DM prefers state C $\Leftrightarrow w^C > w^A$

Higher wealth means higher utility (since more money is always better).

Step 2: $w^C > w^A \Leftrightarrow u'(w^C) < u'(w^A)$

Why? Because of risk aversion ($u'' < 0$), marginal utility is **declining**.

If and only if (iff):
$w^C > w^A \quad \Leftrightarrow \quad u'(w^C) < u'(w^A)$
Since u' is a decreasing function, higher wealth means lower marginal utility.

Step 3: Connect to the FOC

From part (b), at an interior optimum:

$$\frac{u'(w^C)}{u'(w^A)} = \frac{r_1^A - r_2^A}{r_2^C - r_1^C}$$

Rearranging:

$$u'(w^C) = u'(w^A) \cdot \frac{r_1^A - r_2^A}{r_2^C - r_1^C}$$

Question: When is $u'(w^C) < u'(w^A)$?

Answer: When the multiplier is less than 1:

$$\frac{r_1^A - r_2^A}{r_2^C - r_1^C} < 1$$

This happens when:

$$r_1^A - r_2^A < r_2^C - r_1^C$$

Rearranging:

$$r_1^A + r_1^C < r_2^A + r_2^C$$

Interpretation: State C is preferred when the combined returns from asset 2 exceed the combined returns from asset 1 across states A and C.

Common Confusion: Why Less Than 1, Not Greater Than 1?

IMPORTANT: Students often get the inequality direction backwards!

What we want: $u'(w^C) < u'(w^A)$ (state C has lower marginal utility)

Step 1: Divide both sides by $u'(w^A)$ (which is positive):

$$\frac{u'(w^C)}{u'(w^A)} < 1$$

Step 2: From the FOC equation (*):

$$\frac{u'(w^C)}{u'(w^A)} = \frac{r_1^A - r_2^A}{r_2^C - r_1^C}$$

Step 3: Substitute:

$$\frac{r_1^A - r_2^A}{r_2^C - r_1^C} < 1$$

So we need the ratio to be **LESS than 1**, not greater than 1!

Why the confusion?

You might think: "higher wealth should mean greater marginal utility" but that's WRONG!

Remember:

- Higher wealth $\Rightarrow$ **lower** marginal utility (risk aversion!)
- If $w^C > w^A$ (prefer state C because richer)
- Then $u'(w^C) < u'(w^A)$ (lower marginal utility in richer state)
- So the ratio $\frac{u'(w^C)}{u'(w^A)} < 1$ (numerator smaller than denominator)

Bottom line:

$$u'(w^C) < u'(w^A) \iff \frac{u'(w^C)}{u'(w^A)} < 1 \iff \frac{r_1^A - r_2^A}{r_2^C - r_1^C} < 1$$

All three inequalities use **less than (<)**, not greater than!

Concrete Example

Suppose:

- $w_0 = 1000$, $x = 50$
- State A: $r_1^A = 0.1$, $r_2^A = 0.05$ (asset 1 is better)
- State C: $r_1^C = 0.02$, $r_2^C = 0.12$ (asset 2 is better)

Calculate Wealth:

State A:

$$w^A = 1000 + 0.1(50) + 0.05(50) = 1000 + 5 + 2.5 = 1007.5$$

State C:

$$w^C = 1000 + 0.02(50) + 0.12(50) = 1000 + 1 + 6 = 1007$$

In this case: $w^A > w^C$, so the DM prefers state A.

Calculate Marginal Utilities:

If $u(w) = \sqrt{w}$, then $u'(w) = \frac{1}{2\sqrt{w}}$.

State A:

$$u'(w^A) = u'(1007.5) = \frac{1}{2\sqrt{1007.5}} \approx 0.01576$$

State C:

$$u'(w^C) = u'(1007) = \frac{1}{2\sqrt{1007}} \approx 0.01577$$

Notice: $w^A > w^C$ and $u'(w^A) < u'(w^C)$ ✓

This confirms: higher wealth $\Rightarrow$ lower marginal utility.

Why Does This Matter?

The FOC equation (*) balances:

- **LHS:** $\frac{u'(w^C)}{u'(w^A)}$ = ratio of marginal utilities
- **RHS:** $\frac{r_1^A - r_2^A}{r_2^C - r_1^C}$ = ratio of return differences

Interpretation:

The DM adjusts x (investment in asset 1) until:

- The marginal benefit of investing more in asset 1 (captured by return differences)
- Equals the marginal cost in terms of utility (captured by the ratio of marginal utilities)

Risk aversion (u' declining) ensures that:

- States with higher wealth have lower marginal utility
- The DM values an extra dollar more in low-wealth states
- This affects how they balance risk across states

Key Formulas Summary

Concept	Formula/Definition
Marginal utility	$u'(w) = \frac{du}{dw} > 0$
Risk aversion	$u''(w) = \frac{d^2u}{dw^2} < 0$
Declining marginal utility	$u''(w) < 0 \Rightarrow u'$ is decreasing
Prefer state C	$w^C > w^A \Leftrightarrow u'(w^C) < u'(w^A)$
FOC from part (b)	$\dfrac{u'(w^C)}{u'(w^A)} = \dfrac{r_1^A - r_2^A}{r_2^C - r_1^C}$

Common Mistakes

1. **Confusing u' with u**: u' is the *derivative* of utility, not utility itself.

2. **Thinking $u' < 0$**: No! $u' > 0$ always (more money is better). It's $u'' < 0$ that defines risk aversion.

3. **Forgetting the direction**: Risk aversion means u' is *decreasing*, so higher wealth $\Rightarrow$ lower u'.

4. **Not connecting to risk aversion**: The key insight is that risk aversion ($u'' < 0$) is what makes the "iff" statement work.

Key Takeaways

1. u' = **marginal utility** (how much utility per extra dollar)

2. $u'' < 0$ = **risk aversion** (diminishing marginal utility)

3. **Risk aversion makes u' declining:** Higher wealth $\Rightarrow$ lower u'

4. **Prefer state C when** $w^C > w^A$, which by risk aversion means $u'(w^C) < u'(w^A)$

5. **The FOC connects return differences to marginal utility ratios** to determine optimal investment